



Dilshan Narankotuwa

Full Stack Developer

-  narankotuwadilshan@gmail.com
-  0783899883
-  Kandana, Sri Lanka
-  <https://www.linkedin.com/in/dilshan-narankotuwa>
-  <https://github.com/DilshanNarankotuwa>
-  <https://personal-portfolio-zeta-coral.vercel.app>

ABOUT ME

Full Stack Developer specializing in building scalable web applications using React, Node.js, and PostgreSQL. Experienced in designing RESTful APIs, normalized database schemas, and constraint-based validation systems. Strong focus on clean architecture, performance optimization, and production-ready deployment within Linux environments.

TECHNICAL SKILLS

Languages

JavaScript, TypeScript,
Python, R

Frontend

React, Next.js, Tailwind
CSS, Responsive Design

Backend

Node.js, Express.js,
RESTful API Development

Database

PostgreSQL, Prisma ORM

Testing

Jest, Jasmine

Tools & DevOps

Git, Docker, Linux, AWS,
Figma

PROJECTS

GigaHz Compatibility-Based PC Builder

Role: Full Stack Developer

Skills: React, Node.js, PostgreSQL, Prisma

Developed a full-stack PC-building platform with real-time hardware compatibility validation, preventing invalid configurations through CPU socket, RAM type, and power consumption constraints. Architected RESTful APIs and designed a normalized PostgreSQL schema to support dynamic product filtering and build management. Implemented constraint-based validation logic similar to configuration engines used in modern e-commerce systems, emphasizing scalable backend structure and production-ready deployment.

GitHub: <https://github.com/DilshanNarankotuwa/GigaHz>

Live: <https://gigahz-demo.vercel.app>

Galaxy BRIDGE - Molecular Docking & Drug Discovery Pipeline

Role: Computational Research Developer

Skills: Python, Galaxy Tools, Gaussian, Linux, HPC

Designed and automated a molecular docking pipeline targeting dengue NS2B-NS3 protease, optimizing 572 ligand molecules using PM6 geometry optimization. Integrated RMSD statistical validation and visualization workflows while executing large-scale simulations on HPC infrastructure. Focused on reproducible workflow automation and scalable computational execution within HPC environments.

Live: https://galaxy-compchem.ilifu.ac.za/root?tool_id=docking_tool

WORK EXPERIENCE

08/2025 – 11/2025

Volunteer Research Assistant

- Contributed to computational research projects involving molecular modeling and data analysis.
- Assisted in simulation workflows and scientific data validation to ensure reproducibility.

01/2021 – 08/2021

Data Entry & Records Assistant

- Managed high-volume financial datasets with strong accuracy and integrity controls.
- Verified and structured institutional records for compliance and reporting.

01/2020 – 01/2021

Banking Operations Trainee

- Supported structured banking operations and transaction workflows.
- Coordinated documentation processes ensuring regulatory compliance.

EDUCATION

09/2021 – 10/2025

University of Sri Jayewardenepura,

Bachelor of Science (Hons) in Chemistry

- First two years: Intensive foundation in **Mathematics, Statistics, and Chemistry**.
- Final years: Specialized in **Chemistry (Honours)**.
- Focus areas: Computational chemistry, data analysis, molecular modeling.

2019

De Mazenod College - Kandana, Sri Lanka,

G.C.E. Advanced Level Examination

Combined Mathematics, Physics, Chemistry